FOR THE HEMI TEAM

Hemi BTC wallet

Roadmap, cost of each part, and the support the build needs. Payment is in HEMI.

This is an unofficial community wallet: a Bitcoin address and a Hemi address in the same app, unlock with a passkey, and the Hemi tunnel. The PoP page is a remote for popmd on the builder's own machine. The app does not mine, and the phone never holds the miner key. The finished build is its own line, and that line lists what is already in the app.

Price

Each line is a fixed amount in HEMI for that slice. The dollar column is a reference only, using a traded HEMI quote of $0.006 on 2 October 2026. The amount owed is the HEMI column. That quote is the market price of the traded token, not a measurement of PoP payouts. The When column is how long that line takes to build. The lines overlap, so the weeks do not add up to one sequence.

Line	When	HEMI	USD at $0.006
Delivered, already built — $2,400 Passkey wallet. Bitcoin address and Hemi address. Send and receive. Live Bitcoin, Hemi ETH, and hemiBTC. Tunnel screen, both directions. Activity list. Watch-only PoP screen. Debug Android APK 1.1.	now	400,000	$2,400
UI After first review	2 weeks	100,000	$600
v0 Connected status	3 weeks	150,000	$900
v1 Track mint and payout	5 weeks after v0	250,000	$1,500
v2 Miner remote	5 weeks after sync	250,000	$1,500
v3 Home, alerts, passkey	3 weeks after v2	150,000	$900
UI Closing review	2 weeks after v3	100,000	$600
Total		**1,400,000**	**$8,400**

The $2,400 row is work already finished, including both tunnel screens. v1 does not build those screens again. There are two UI passes. Each one is a single written list of layout, labels, and flow on screens that already exist. A new feature is not part of either pass. If the HEMI price has moved a long way before this sheet is accepted, either side can ask to reset the HEMI amounts once, before the first payment. After that, each line stays at the HEMI figure above.

What the $2,400 already includes

400,000 HEMI, shown as $2,400 in the table. This is already in the web app and in a debug Android build, version 1.1. These pieces are in the build:

- Passkey wallet. One recovery phrase. The wallet is saved and opened with a passkey. A password is optional, and only as a second step. On the phone, unlock uses the device biometric store. Several wallets can be saved on the device, each under a name.
- Bitcoin address and Hemi address. Both come from that phrase and both are shown in the app. Each has a receive screen. Bitcoin, ETH on Hemi, and hemiBTC can each be sent from the wallet.
- Live balances. Bitcoin, ETH on Hemi (the gas on that chain), and hemiBTC. The bitcoin total also shows a dollar figure when a price is available.

- Tunnel, both directions. Deposit and withdraw are both on the tunnel screen. A deposit quotes the vault fee, is built to pay the live vault, and writes the Hemi address into the Bitcoin memo. A withdrawal quotes the fee and is built to make the one vault call. A mainnet checkbox has to be on before either broadcast. Activity keeps each move and, after a deposit broadcast, has a mint button.
- Activity. Tunnel moves made on this device stay listed on the device. Bitcoin and hemiBTC transfers for the two addresses are listed from the chains.
- PoP screen, watch only. The next 25-block keystone, the public Bitcoin fee, recent proofs, and the HEMI balance of the address on screen. This screen does not connect to a miner and does not spend bitcoin.
- Android build. Debug APK version 1.1, the same app, so the current build can be installed on a phone.

Not included in the $2,400. A full mainnet deposit through minted hemiBTC, and a full withdrawal through paid Bitcoin, have not been completed. Reading the state of a deposit that was already sent is v0. Completing a new deposit through mint, and a withdrawal through payout or the 12-hour grace, is v1. The PoP screen does not yet store a Tailscale address. The read-only connection is v0. Start, stop, the fee, and the last proof's block are v2.

Accepted when the team can open the current build, unlock with a passkey, and see the Bitcoin address, the Hemi address, the live balances, send and receive, the tunnel screen in both directions, the activity list, and the watch-only PoP screen.

UI · After the first review

100,000 HEMI · $600 at the reference price. About 2 weeks after one written list of notes on the delivered build. The new disk is not required. This pass can run beside v0.

The team reviews what is already built and sends one list. This line changes layout, labels, the order of actions, empty states, and error text on the wallet, receive, send, tunnel, activity, and the watch-only PoP screen. The same build is updated on the web and in the Android APK. Two weeks is enough for one list on these screens. A new flow, or a second list, is not this line. If that list is large, v0 slips by the same amount, because one person is building both.

Accepted when the agreed changes from that one list are in a build the team can open on the web and in the Android build.

The phone is a remote

popmd on the Ubuntu machine is the miner. The app shows its status and, from v2, sends start, stop, and a fee change. A small service is installed beside popmd and is the only thing the phone talks to. It is an ordinary program with one config file and a service unit. Someone can install it on the same machine as popmd. It reads popmd's health page on 127.0.0.1, and it is allowed to start, stop, and restart only that service.

The miner key stays in popmd's own config. The service cannot read that file, and the PoP page has no field for a key. popmd's health page stays on localhost. The service listens on Tailscale, with a rule that allows only the owner's phone, plus a token checked on every call. Nothing is opened on the public internet.

The PoP page has two fields: the service's Tailscale address, such as https://popminer.tailnet.ts.net, and the token set when the service was installed. The phone remembers both after the first successful check. The Tailscale app has to be connected on the phone. The address points at the service, not at popmd. Start is a separate confirm, because starting spends bitcoin. A fee change restarts popmd, because popmd reads the fee only at startup.

v0 is the read-only service: whether popmd is reachable, and whether Bitcoin and Hemi look connected. popmd's health page does not include the fee or the last proof, so those are v2, with start, stop, and the fee restart. The 2 TB Bitcoin index and the Hemi node have to be on that machine before a live reading. The service does not replace them. Until they are synced, v0 is demonstrated against a stand-in health page, so this line is not waiting on the disk.

v0 · Connected status

150,000 HEMI · $900 at the reference price. About 3 weeks from acceptance. The miner machine is not required.

For a deposit already broadcast, Activity reads Bitcoin and Hemi and shows sent, confirming, minted, or failed, using the vault's own confirmation count. The state is still there after the app is closed and opened. The PoP page stores the Tailscale address and the token, and shows whether the service can reach popmd and whether Bitcoin and Hemi look connected. The fee and the block of the last proof are v2. If the miner is not synced yet, acceptance uses a stand-in health page on the builder's computer.

Accepted when a deposit that was already sent still shows sent, confirming, minted, or failed after the app is reopened, and the PoP page shows health from the service or from that stand-in.

v1 · Track mint and payout

250,000 HEMI · $1,500 at the reference price. About 5 weeks after v0. The disk is not required. The deposit and withdraw screens are already built.

This is the first completed mainnet round trip, so the five weeks include waiting on Bitcoin confirmations and on the vault payout, which can take up to 12 hours, plus a retry if the first attempt fails. A new deposit is followed until hemiBTC is minted. The screen shows the vault's confirmation count, then the mint, and it shows that same state after the app was closed. A withdrawal already makes the one vault call. The screen then shows waiting on the operator, paid in Bitcoin, or past the 12-hour grace, including after the app was closed. A rejected send, a mint that does not arrive, or a payout that does not arrive stays on the screen with the reason. The builder supplies the small amount of bitcoin for the test.

Accepted when one small mainnet deposit is followed until hemiBTC is minted, and one withdraw above the vault minimum is followed until Bitcoin arrives or the grace passes, or the screen shows the real reason it did not.

v2 · Miner remote

250,000 HEMI · $1,500 at the reference price. About 5 weeks after popmd has finished syncing. Buying the machine and the sync are not inside these 5 weeks.

Start, stop, and the fee cap go through the same service. Start asks for a confirm, because it spends bitcoin. The fee is kept in a file that does not contain the miner key. A new fee is written there and popmd is restarted so it picks the fee up. Each proof shows the Bitcoin block it landed in, and whether that block is inside the nine-block pay window. The app reads that from Bitcoin, because popmd’s health page does not report it. A rewards line shows HEMI credited to the miner address once the payout contract is available. The night test does not depend on that contract.

Accepted when the builder runs one night on their own machine, started and stopped from the phone, the fee cap holds, and each proof shows its Bitcoin block.

v3 · Home, alerts, passkey

150,000 HEMI · $900 at the reference price. About 3 weeks after v2.

The home screen shows the tunnel and the miner remote together. The app checks for four events: a deposit minted, a withdrawal paid, a proof outside the pay window, and the miner paused because the fee is above the cap. The check runs when the app is open, and the phone can raise a notification when the system allows the app to keep checking. An app that has been force-quit does not page the phone. Spending bitcoin, and starting the miner, ask for the passkey again.

Accepted when each of those four events shows an alert the next time the app checks, on a real event, and a spend or a miner start asks for the passkey.

UI · Closing review

100,000 HEMI · $600 at the reference price. About 2 weeks after v3, once the team has used the finished app and sent one list.

One list of layout, labels, order, empty states, and error text across home, tunnel, activity, the PoP remote, alerts, and unlock. The web build and the Android build are both updated. Two weeks covers that one list. It does not add a feature past v3, and it does not cover a second list.

Accepted when the agreed changes from that one list are in a build the team can open on the web and in the Android build.

Support needed from Hemi

Four things from the team. The builder covers the test bitcoin, the miner computer, the miner key, and the small service beside popmd.

- Keystones. An op-geth endpoint that answers kss_getLatestKeystones, or the document for the endpoint miners are expected to use. The public Hemi RPC does not serve that call. Current popmd also needs its own Bitcoin index (tbcd). The older public BFG address is the previous miner.
- Payout contract. The PoP payout contract address, its interface, and whether a keystone still pays the published 100 HEMI before the split. That figure is from the August 2025 miner notes. It has not been read off a live contract, because the contract address was not found in public search.
- A check of the tunnel already wired. Factory 0xEAcA824F46c000fB89403846Bb57e6b913321081, vault index 8, vault 0x58f7b8d7a7291AAEcE0febb39aa4e877387e61E4, hemiBTC 0xAA40c0c7644e0b2B224509571e10ad20d9C4ef28, HEMI 0x99e3dE3817F6081B2568208337ef83295b7f591D, chain 43111. Contract minimum is 4 Bitcoin confirmations. The grace period on a withdrawal is 12 hours. Fees on the vault are 20 bps with a floor of 500 sats.
- Name. A yes or no on staying an unofficial wallet. The app keeps the Hemi orange and the name Hemi BTC, and it is described as unofficial unless the team says otherwise.

Order of work

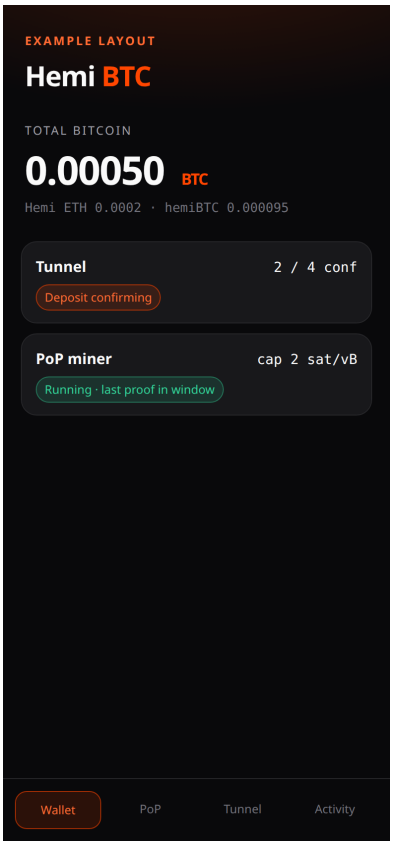
The delivered build can be reviewed now. v0 starts when this sheet is accepted and does not wait on the miner machine. The first UI pass starts when notes arrive and runs beside v0. v1 starts when v0 is accepted, so the tunnel follow-through is about 8 weeks from acceptance. v2 waits on a separate Ubuntu machine with a 2 TB NVMe and 32 GB of RAM, running popmd and the small service. That machine is not bought yet. The builder expects to order it about a week after this sheet. The Bitcoin index was already over 1 TB in June 2025, so the sync takes longer than the purchase and can take several weeks. v2’s five weeks start after that sync, not after acceptance. v3 is about 3 weeks after v2, and the closing UI pass is about 2 weeks after v3.

If the machine is ordered on that week and the sync takes a few weeks, the closing pass ends about three to four months after acceptance. A later machine moves v2, v3, and the closing pass by the same delay. v0, v1, and the first UI pass do not move.

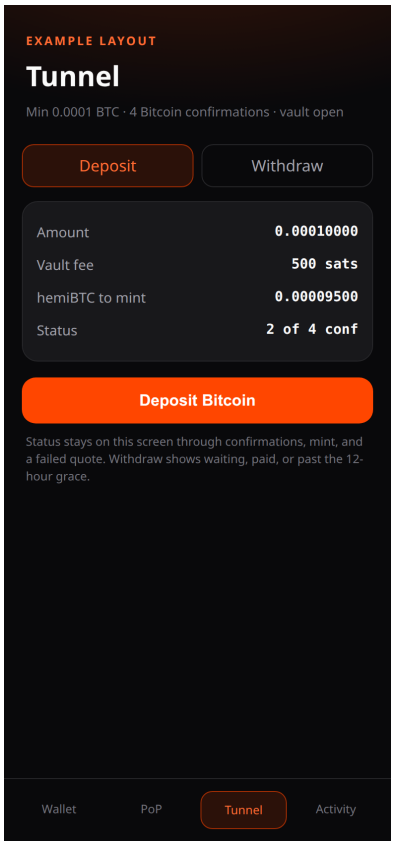
The delivered line is payable when the team has reviewed the current build. Each later line, including each UI pass, is payable when its acceptance line is met. There is no retainer. The miner computer is outside this price.

Frames

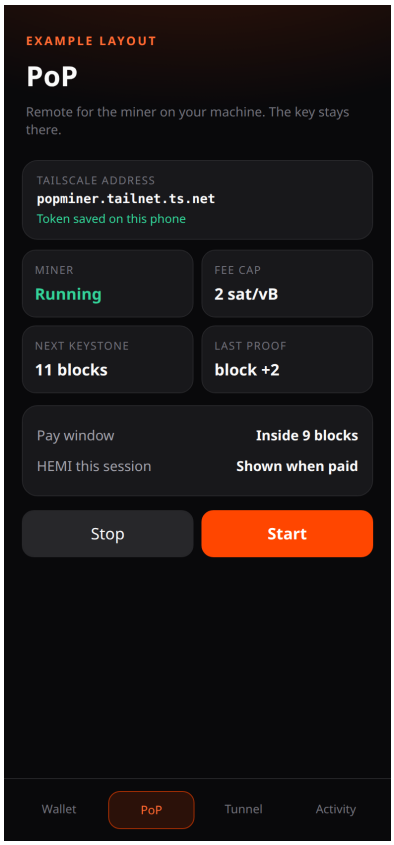
Target screens for later milestones. The PoP frame shows the remote: a saved Tailscale address, the token kept on the phone, and start and stop for the miner on the other machine. The figures are a sample layout. They are not a live balance and not a count of HEMI earned.



Home. Tunnel and the miner remote together.



Tunnel. Fee, mint amount, and confirmations.



PoP. Tailscale address, token, start and stop.